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Homework 1 (Due: March 20th)

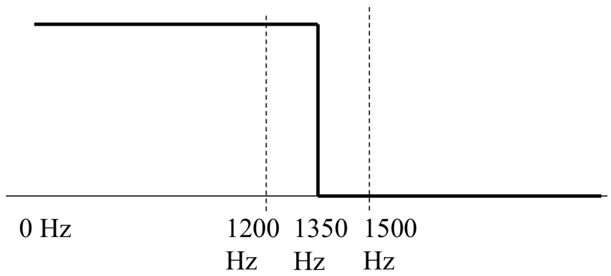
(1) Design a Mini-max **lowpass** FIR filter such that (40 scores)

① Filter length = 17, ② Sampling frequency f_s = 6000Hz,

③ Pass Band 0~1200Hz ④ Transition band: 1200~1500 Hz,

⑤ Weighting function: $W(F)$ = 1 for passband, $W(F)$ = 0.6 for stop band .

⑥ Set Δ = 0.0001 in Step 5.



※ The code should be handed out by NTUCool, too.

Show (a) the frequency response, (b) the impulse response $h[n]$, and (c) the maximal error for each iteration.

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Iteration 0 max_err is : 0.38270
Iteration 1 max_err is : 0.26679
Iteration 2 max_err is : 0.34017
Iteration 3 max_err is : 0.48625
Iteration 4 max_err is : 0.08491
Iteration 5 max_err is : 0.07124
Iteration 6 max_err is : 0.07121
Iteration 7 max_err is : 0.07121
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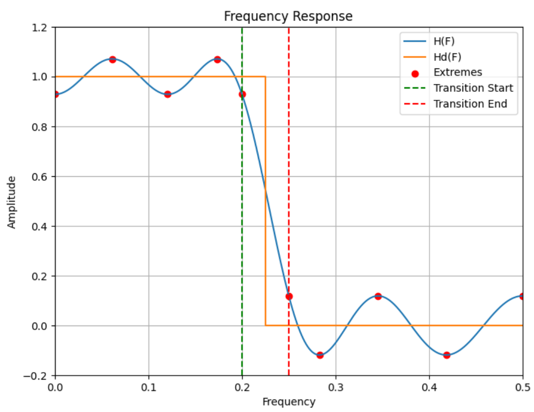
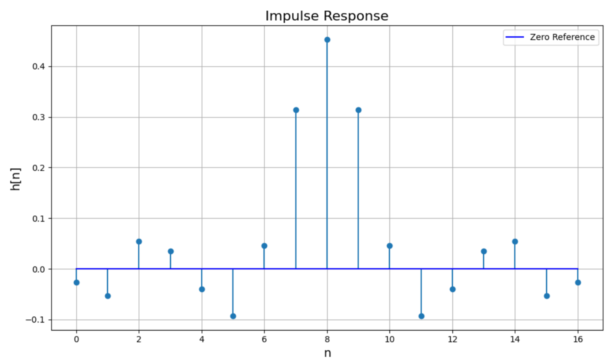
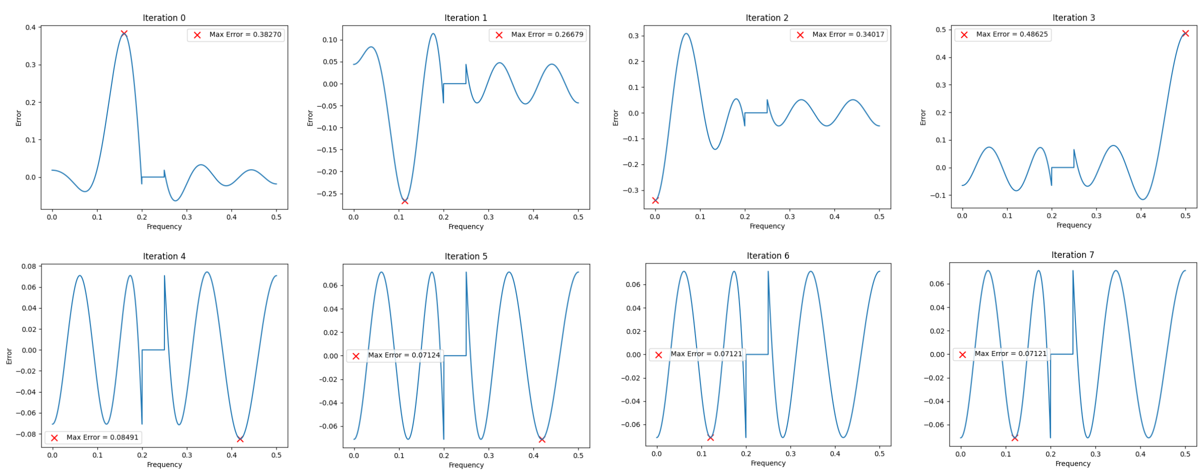
(a)



(b)



(c)



(2) How do we implement $y[n] = x[n] *(0.8^n u[n] + 0.5^n u[n])$ efficiently where * means convolution and $u[n]$ is the unit step function? (10 scores)

$$y[n] = x[n] * (0.8^n u[n] + 0.5^n u[n])$$

$$\xrightarrow{Z} \quad Y(z) = X(z)\,\underline{H(z)}$$

$$H(z) = \sum_{n\to\infty}^{\infty} h(n)\, z^{-n}$$

$$= \sum_{n=0}^{\infty} 0.8^n z^{-n} + \sum_{n=0}^{\infty} 0.5^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (0.8 z^{-1})^n + \sum_{n=0}^{\infty} (0.5 z^{-1})^n$$

$$= \frac{1}{1-\frac{0.8}{z}} + \frac{1}{1-\frac{0.5}{z}}$$

* 無窮等比級數和 $\frac{a(1-r^n)}{1-r}$

$$= X(z)\cdot\left(\frac{1}{1-\frac{0.8}{z}} + \frac{1}{1-\frac{0.5}{z}}\right)$$

$$= X(z)\cdot \frac{2-1.3z^{-1}}{(1-0.8z^{-1})(1-0.5z^{-1})}$$

$$(1-1.3z^{-1}+0.4z^{-2})\,Y(z) = X(z)\,(2-1.3z^{-1})$$

$$\xrightarrow{z^{-1}} \quad y[n] - 1.3y[n-1] + 0.4y[n-2] = 2x[n] - 1.3x[n-1]$$

$$y[n] = 2x[n] - 1.3x[n-1] + 1.3y[n-1] - 0.4y[n-2]_{\#}$$

(3) (a) What are the two main advantages of the Fourier transform (FT)? (b) What are the two main problems to implement the FT? (10 scores)

(a) 1. spectrum analysis

可以頻譜分析！

2. convolution $\longrightarrow$ multiplication

可以把捲積變成乘法！

$y(t) = x(t) * h(t) \longrightarrow Y(f) = X(f)H(f)$

$= \int x(t-\tau)h(\tau)\,d\tau$

(b) 1. not real operation

複數運算較麻煩！

2. irrational number multiplication

* $e^{-j2\pi ft} = \cos(2\pi ft) - j\sin(2\pi ft)$

三角大多都是無理數

(4) Suppose that $x[n] = y(0.002n)$ and the length of $x[n]$ is 2000. If $X[m]$ is the FFT of $x[n]$, which frequencies do (a) $X[200]$ and (b) $X[1600]$ correspond to? (10 scores)

$t = n\Delta t$ $\qquad$ $N = 2000$

$\Rightarrow \Delta t = 0.002$, $f_s = \frac{1}{\Delta t} = 500$

(a) $X[200]$, $m = 200$

$$f = m \times \frac{f_s}{N} = 200 \times \frac{500}{2000} = 50\text{ Hz}_{\#}$$

(b) $X[1600]$, $m = 1600$

$$f = m \times \frac{f_s}{N} = 1600 \times \frac{500}{2000} = 400\text{ Hz} > f_0 = 250\text{ Hz}$$

(最大可分析頻率)

$$\Rightarrow f = 400 - f_s = -100\text{ Hz}_{\#}$$

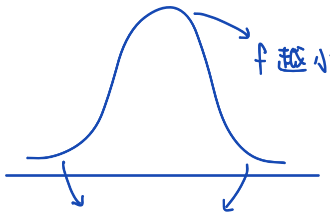
(5) Why (a) the step invariance method and (b) the bilinear transform can reduce or avoid the aliasing effect in IIR filter design? (10 scores)

(a) step invariance ⇒ reduce aliasing effect，一開始就將 input $h_a(t)$ 與 step function 做 convolution，

$$h_{a,u}(t) = h_a(t) * u(t) = \int_{-\infty}^{\infty} h_a(\tau)u(t-\tau)\,d\tau = \int_{-\infty}^{t} h_a(\tau)\,d\tau$$

(其實就是對 $h_a(t)$ 做積分)

$$\Rightarrow \quad H_{a,u}(f) = \frac{H_a(f)}{j2\pi f}$$



f 越小，H_{au} 就變大

f 越大，H_{au} 就變小

⇒ 高頻成分被壓低，aliasing effect 降低！ ※

(b) bilinear transform ⇒ avoid aliasing effect

強制把無限長的類比 filter mapping 到 $-\frac{f_s}{2} \sim \frac{f_s}{2}$ 之間

$$f_{new} = \frac{f_s}{\pi}\,\text{atan}\left(\frac{2\pi}{c} f_{old}\right)$$ ⇒ 以防止產生 aliasing effect！ ※

(6) (a) Which of the following filters are usually even? (b) Which of the following filters are usually odd? (i) Notch filter; (ii) highpass filter; (iii) edge detector; (iv) integral; (v) differentiation 4 times; (vi) particle filter; (vii) matched filter. (10 scores)

(a) even：(i) Notch filter (ii) highpass filter (v) differentiation 4 times

(b) odd：(iii) edge detector (iv) integral

(vii) matched filter ⇓ 由參考信號來決定是 odd (與參考信號一致) 或 even

(vi) particle filter ⇓ 本身並不具備對稱性 ※

(7) Use the MSE method to design the 7-point FIR filter that approximates the low ~~high~~pass filter of $H_d(F) = 1$ for $|F| < 0.25$ and $H_d(F) = 0$ for $0.25 < |F| < 0.5$. (15 scores)

$$\begin{cases} H_d(F) = 1 \text{ for } |F| < 0.25 \\ H_d(F) = 0 \text{ for } 0.25 < |F| < 0.5 \end{cases}$$

$$S[0] = \int_{-\frac{1}{2}}^{\frac{1}{2}} H_d(F)\,dF = \int_{-\frac{1}{4}}^{\frac{1}{4}} 1\,dF = 0.5$$

$$S[n] = 2\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos(2\pi nF) H_d(F)\,dF = 2\int_{-\frac{1}{4}}^{\frac{1}{4}} \cos(2\pi nF)\,dF$$

$$= 2 \times \frac{1}{2\pi n}\left(\sin(2\pi nF)\Big|_{-\frac{1}{4}}^{\frac{1}{4}}\right)$$

$$= \frac{1}{\pi n}\left(\sin\left(\frac{\pi n}{2}\right) + \sin\left(\frac{\pi n}{2}\right)\right) = \frac{2}{\pi n}\sin\left(\frac{\pi n}{2}\right)$$

$S[1] \approx 0.63662$

$S[2] \approx 0$

$S[3] \approx -0.21221$

h[0]	h[1]	h[2]	h[3]	h[4]	h[5]	h[6]
$= \frac{S[3]}{2}$	$= \frac{S[2]}{2}$	$= \frac{S[1]}{2}$	$= S[0]$	$= \frac{S[1]}{2}$	$= \frac{S[2]}{2}$	$= \frac{S[3]}{2}$
$= -0.106105$	$= 0$	$= 0.31831$	$= 0.5$	$= 0.31831$	$= 0$	$= -0.106105$

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